

ETL Architecture and Tooling

Pattern

Bronze/silver/gold batch pipeline from source files to analytics-ready tables.

Tools

Python + pandas for ingestion/profiling, SQL/dbt-style marts for warehouse outputs, scheduler for daily orchestration.

Controls

File manifest, source lineage, schema validation, data quality gates, and partitioned gold publication.

Ingestion and Transformation Decisions

Credit snapshots

Append incrementally by reporting_date parsed from filenames; replay late files by replacing affected partitions.

Customer and NPS workbooks

Full refresh dimensions where source is workbook-based; preserve NPS repeated responses and dedupe by submission_id.

Feature automation

Normalize loan_id, join DOB/income/NPS, then derive age_band, income band, days_past_due, and risk_category.

Storage, Quality, and Recovery

Gold outputs

gold_credit_account_snapshot, gold_portfolio_metrics_daily, gold_risk_segment_metrics, gold_credit_nps_customer_view.

Partitioning

Partition by reporting_date; cluster/index by loan_id, risk_category, age_band, and avg_monthly_income_band.

Failure handling

Quarantine malformed files, block gold publication on critical DQ failures, alert stakeholders, and replay corrected files.